

A Problem With Fibonacci Sequence

Tags:

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Description

Fibonacci sequence is well-known in the world. We define Fibonacci sequence as follows: $F(0) = 0$, $F(1) = 1$, $F(n) = F(n-1) + F(n-2)$, $n \geq 2$. It's easy for us to calculate $F(n) \bmod m$. But this time we want to make the problem more difficult. We want to calculate the formula:

$$\sum_{k=0}^n C_n^k F(k), \quad C_n^k \text{ is the combination number.}$$

Input

First line is the testcase T . Following T lines, each line is two integers n, m ($0 \leq n \leq 10^9$, $1 \leq m \leq 30000$).

Output

Output the answer mod m .

Samples

input	Copy
2	
1 30000	
2 30000	
output	Copy
1	
3	

Source

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